MSTAR PRACTICE WORKSHEET

Lesson 8-6: Solve and Graph Inequalities

Grade 6 Mathematics · Standard 6.AT.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Solve: $x + 6 > 14$

- A $x > 8$
- B $x < 8$
- C $x = 8$
- D $x > 20$

Subtract 6 from both sides: $x > 14 - 6 = 8$. Graph: open circle at 8, shade right.

Question 2 SELECTED RESPONSE — CORRECT: B

Solve: $x - 9 \leq 3$

- A $x \leq 6$
- B $x \leq 12$
- C $x \geq 12$
- D $x = 12$

Add 9 to both sides: $x \leq 3 + 9 = 12$. Graph: closed circle at 12, shade left.

Question 3 SELECTED RESPONSE — CORRECT: A

Solve: $x + 10 \geq 25$

- **A** $x \geq 15$
- B $x \leq 15$
- C $x = 15$
- D $x \geq 35$

> Subtract 10 from both sides: $x \geq 25 - 10 = 15$. Graph: closed circle at 15, shade right.

Question 4 SELECTED RESPONSE — CORRECT: B

Solve: $x - 4 < 11$

- A $x < 7$
- **B** $x < 15$
- C $x > 15$
- D $x = 15$

> Add 4 to both sides: $x < 11 + 4 = 15$. Graph: open circle at 15, shade left.

Question 5 SELECTED RESPONSE — CORRECT: A

Is $x = 5$ a solution to $x + 4 > 12$?

- **A** No, because $5 + 4 = 9$, and 9 is not greater than 12
- B Yes, because $5 + 4 = 9$
- C Yes, because $5 > 4$
- D No, because $5 < 12$

> Substitute: $5 + 4 = 9$. Since $9 > 12$ is false, $x = 5$ is NOT a solution.

Question 6 SELECTED RESPONSE — CORRECT: D

Clue 1: Which value is a solution of $x > 7$?

- A 3
- B 5
- C 7
- D 9

A solution makes $x > 7$ true. Substitute each: $9 > 7$ is true, but $7 > 7$, $5 > 7$, and $3 > 7$ are all false. So 9 is the solution that opens the gate.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Solve n – 6 > 9. Which number line shows the solution set?

(A) An open circle at 3 with shading to the right

(B) An open circle at 15 with shading to the right

(C) A closed circle at 15 with shading to the right

(D) An open circle at 15 with shading to the left

> Add 6 to both sides to isolate the variable: n > 9 + 6, so n > 15. The symbol > uses an open circle at 15, and the solutions are the numbers greater than 15, so the shading goes to the right.

- **A:** That subtracts 6 from 9. To undo a subtraction of 6, add it: 9 plus 6.
- **C:** The symbol is greater than, not greater than or equal to, so 15 itself is not a solution.
- **D:** The boundary is right but the direction is not. Values above 15 make the statement true.

Part B — correct answer: A

Which reasoning supports that graph?

(A) Adding 6 to both sides gives n > 15, and > leaves the boundary point unfilled with shading toward the larger numbers.

(B) Subtracting 6 from both sides gives n > 3, and > leaves the boundary point unfilled.

(C) Adding 6 to both sides gives n > 15, and the circle is filled in because 15 makes the inequality true.

(D) Adding a number to both sides reverses the inequality symbol, so the shading points to the left.

> The inverse of subtracting 6 is adding 6, which gives n > 15. Substituting the boundary gives 15 – 6 = 9, and 9 is not greater than 9, so 15 is not a solution and the circle stays open. Substituting 16 gives 16 – 6 = 10, which is greater than 9, so the shading covers the numbers above 15.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Subtracting Away a Coefficient

Tomas solved $3x \le 18$. He wrote: '$3x \le 18$, so $x \le 18 - 3$, which gives $x \le 15$.' He then drew a closed circle at 15 and shaded to the left.

Explain Tomas's misconception and give the correct solution and graph.

Model answer: $x \le 6$, graphed with a closed circle at 6 and shading to the left. In $3x$ the variable is multiplied by 3, so the inverse operation is division: divide both sides by 3 to get $x \le 18 \div 3$, which is $x \le 6$. The symbol $\le$ includes the boundary point, so the circle at 6 is filled in. Tomas's answer fails a check, because $3 \times 15 = 45$, and 45 is not less than or equal to 18. His circle and shading were right in style but sat at the wrong boundary number.

2 points	Student explains that 3 is multiplied by the variable, so both sides must be divided by 3 instead of having 3 subtracted, gives $x \le 6$, and describes a closed circle at 6 with shading to the left.
1 point	Student gives $x \le 6$ but does not explain why division is the inverse operation that is needed.
0 points	Student agrees with Tomas or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A detective has a budget of at most \$100 for supplies. She already spent \$37 on gloves. Write an inequality for the remaining amount r she can spend, solve it, graph it, and name three possible values for r.

Model answer: "At most \$100" puts a cap on the total, so the model is $37 + r \le 100$. Subtracting 37 from both sides gives $r \le 63$. Because "at most" allows the boundary itself, the graph is a closed circle at 63 shaded to the left, and amounts such as \$63, \$40, and \$10 are all possible values of r.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.